Dhruv Patel

Los Angeles, CA | U.S. Citizen | (714) 707-1591 | dhruvpatel97@ucla.edu | LinkedIn | GitHub | Website

Education

University of California, Los Angeles (UCLA) Jun. 2027

B.S. Mathematics of Computation, Data Science Engineering Minor *GPA: 3.77/4.00*

- **Relevant Coursework:** Machine Learning, Deep Learning, Natural Language Processing, Probability, Linear Algebra, AI for Scientific Computing (Graduate), AI for Finance (Graduate), Numerical Methods, Algorithms
- **Honors:** Upsilon Pi Epsilon (CS Honor Society), UCLA Dean's Honors List (3×), Eagle Scout
- **Organizations:** ACM AI, Undergraduate Mathematics Student Association, Poker Group, Cubing Club, Intramural Basketball

Experience

UCLA Mathematical & Statistical Finance Group Los Angeles, CA

Machine Learning Research Assistant *Jun. 2025 – Present*

- Co-authoring a paper with Prof. Mihai Cucuringu forecasting how unevenly daily trading volume spreads across stocks
- Benchmarked 10+ models and tested spectral-GMM clustering for regime detection; augmented HAR cut MSE 28.5% vs AR(1)
- Constructed a market-neutral long-short strategy on these forecasts with 1.2–1.5 Sharpe in US and China OOS backtests
- Developed AlphaMark, an open-source backtester turning forecast signals into PnL reports; 4× speedup via a 5D tensor

Pacific Life Newport Beach, CA

Data Science Intern *Jun. 2026 – Sep. 2026*

- Implemented semantic search over medical records (Snowflake Cortex) to surface underwriting-relevant details
- Added an LLM stage flagging mismatches between records and applicant answers, projected to cut underwriter caseload ∼10%
- Automated tuning of the prompt, model, and similarity threshold with an agent loop to minimize false omission rate
- Deployed an Azure DevOps pipeline tracking per-user Snowflake spend, sending AWS SNS alerts when daily spend tops $50

Smart Circle International Newport Beach, CA

AI Engineer Intern *Jun. 2025 – Jun. 2026*

- Built and tested AT&T churn models (Random Forest, Logistic Regression), reaching 50%+ recall at 85%+ accuracy
- Turned 100K+ sales dispositions into a weekly Tableau efficiency report for Lumen, later adopted across 15+ campaigns
- Engineered a Python pipeline parsing SCORM e-learning content across 10+ formats into structured data for AI workflows
- Created a Snowflake Cortex chatbot for AT&T territory reporting and zip code-level lead replacement recommendations
- Defined office-level metrics to evaluate AI-driven recruiting, informing funding allocation across 100+ offices

Bruin Sports Analytics Los Angeles, CA

Tennis Data Scientist *Oct. 2024 – Jun. 2026*

- Prototyped a PyTorch transformer predicting shot type and court location from sequential rally data
- Produced 20+ serve-tendency visualizations the UCLA D1 Men's Tennis team used to adjust pre-match strategy
- Standardized the footage-to-visualization workflow end to end, enabling adoption across the scouting team

Projects

LOBSTER News Alpha Agent | *Python (pandas, SciPy, NumPy), Anthropic API*

- Designed an LLM agent that turns news headlines and limit order book flow into trade decisions across 2,879 events
- Beat a lexicon baseline on inference-heavy TSLA headlines (IC +0.28, $p < 0.001$), the only subgroup surviving BH-FDR

In-Context Operator Networks for Solving PDEs | *Python (PyTorch, SciPy, Sentence Transformers)*

- Trained a 2.21M-parameter transformer that solves 1D viscous Burgers from in-context examples, based on ICON-LM
- Cut held-out error from 22% to 15% with 3 prompt examples; text captions alone cut zero-shot error up to 27%

EEG Motor Imagery Classification | *Python (PyTorch, NumPy, SciPy), Google Cloud Platform*

- Compared 5 architectures (EEGNet, TCN, CNN+LSTM, CNN+GRU, Transformer) with 36-fold leave-one-subject-out CV
- Showed compact models (<4K params) generalize best: TCN hit 45.2% cross-subject accuracy vs 41.1% for a 200K-param LSTM

Skills

Languages: Python, SQL, C++, R, Java, C, JavaScript, HTML/CSS, LaTeX

Libraries: PyTorch, pandas, NumPy, Scikit-Learn, SciPy, Matplotlib, Seaborn, XGBoost, LightGBM, Optuna, hmmlearn

Tools: Snowflake, AWS, Docker, Git, Linux, Azure DevOps, Google Cloud Platform, JupyterLab, Tableau, Excel